IDENTIFICATION

Forensic Medicine | 3rd Year MBBS

Hi Sammy!
Identification = one of
the BIGGEST FMT topics!
Fingerprints are key!

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Corpus Delicti + Methods

Corpus delicti = body of the crime. Identity must be established!

CORPUS DELICTI

"The body or substance of the crime — proving that a crime has actually been committed." — Reddy

Data Used for Identification

Category	Parameters
Physical	Sex, age, race, stature, build, complexion, hair, eyes
Anthropometric	Bertillon system — 11 body measurements (obsolete now)
Fingerprints	Most reliable, unique, permanent, classifiable (Galton)
DNA profiling	Gold standard for identity — unique except identical twins
Dental	Most resistant to destruction; Gustafson's method for age
Scars/Tattoos	Occupational marks, surgical scars, vaccination marks
Skeletal	Bones for sex, age, race, stature estimation
Clothing/Jewellery	Personal belongings — circumstantial evidence

Identification — Living vs Dead

LIVING PERSON

- Photographs, fingerprints
- Scars, tattoos, deformities
- Gait, voice, handwriting
- Occupation marks
- Anthropometric measurements

DEAD / DECOMPOSED

- All methods for living +
- Dental records comparison
- DNA profiling from remains
- Skeletal examination
- Superimposition technique

Skull-Photo Superimposition

Skull photograph overlaid on ante-mortem photo using video camera.

16 anatomical landmarks matched. Used in Ruxton case (1935) — first use.

Fingerprints = most reliable ID method! Galton's classification!

Fingerprints — Dactylography

Appear:	12-16 weeks intrauterine life; permanent throughout life
Destroyed:	Only by destruction of dermis (deep burns, strong acids)
Unique:	No two identical fingerprints ever found (not even identical twins)
Basis:	Friction ridges on fingertips — genetically determined but unique

Galton's Classification of Fingerprints

Type	Frequency	Pattern Description
Loop	67% (most common)	Ridges enter from one side, recurve, exit same side
Whorl	25%	Circular/spiral pattern; at least 2 deltas
Arch	7%	Ridges run from one side to other; no delta (plain arch)
Composite	1% (rarest)	Combination of 2+ patterns (twinned loop, accidental)

Loop subtypes: Ulnar loop (opens towards ulna — most common) |
Radial loop (opens towards radius — less common)

DNA Profiling (DNA Fingerprinting)

Feature	Detail
Developed by	Sir Alec Jeffreys (1984, UK)
India pioneer	Dr. Lalji Singh (Centre for DNA Fingerprinting, Hyderabad)
Method	PCR + STR (Short Tandem Repeat) analysis — current standard
Uniqueness	1 in billions probability (except identical twins)
Source material	Blood, semen, saliva, hair root, bone, teeth
Uses	Paternity, rape cases, mass disasters, decomposed bodies
Indian law	DNA Technology (Use & Application) Regulation Act 2019

★ Remember!
Fingerprints = most reliable routine method | DNA = gold standard (but expensive)

Gustafson's method = guaranteed question! 6 parameters for age!

Age Estimation — Quick Overview

Method	Details
Dentition	Deciduous: 6mo-2yr | Permanent: 6-25yr | 3rd molar = 17-25yr
Ossification	Appearance & fusion of epiphyses — most accurate in children
Secondary sexual	Puberty changes: breast, pubic hair, voice (Tanner staging)
Radiological	X-ray wrist, pelvis, clavicle (medial end fuses last: 25yr)
Histological	Bone/tooth microscopy — less accurate, used for adults

Gustafson's Method (1950) — 6 Parameters

#	Parameter	Change with Age
A	Attrition	Wearing of occlusal surface — increases with age
P	Periodontosis	Recession of gum/periodontal attachment
S	Secondary dentin	Deposition inside pulp cavity — reduces pulp size
C	Cementum	Increased deposition of cementum on root
R	Root resorption	Resorption of root apex — transparent/translucent
T	Root transparency	Sclerosis of dentinal tubules — root becomes transparent

★ Mnemonic: A-P-S-C-R-T — **A Perfectly Smart Cat Runs Twice! (Attrition, Periodontosis, Secondary dentin,** Cementum, Root resorption, Transparency) — each scored 0-3, total = age formula

Dental Identification — Key Points

- Teeth = most resistant structures in the body (survive fire, decomposition)
- Dental records comparison = most reliable for mass disaster victim identification
- Bite marks can be used as evidence (compared with suspect's dental cast)
- Rugoscopy (palatal rugae pattern) — unique to individual, used for ID

Identification is a guaranteed LAQ topic! You've GOT this Sammy!

Most Repeated LAQs (10-12 marks)

1. Describe methods of identification of a dead body
2. Classify fingerprints (Galton). Discuss medicolegal importance
3. Describe Gustafson's method of age estimation from teeth
4. Write about DNA profiling — method, uses, and medicolegal importance
5. Discuss age estimation methods from birth to 25 years

Most Repeated SAQs (5-6 marks)

1. Galton's classification of fingerprints
2. Gustafson's 6 parameters (with mnemonic)
3. Skull-photo superimposition technique
4. Corpus delicti — definition and importance
5. DNA profiling vs fingerprints — comparison
6. Rugoscopy and cheiloscopy in identification

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Most reliable method of ID	Fingerprints (routine) / DNA (absolute)
Most common fingerprint type	Loop (67%)
Rarest fingerprint type	Composite (1%)
DNA profiling pioneer	Sir Alec Jeffreys (1984)
Indian DNA pioneer	Dr. Lalji Singh (CDFD, Hyderabad)
Gustafson's method — parameters	6 (APSCRT) — scored 0-3 each
Most resistant body structure	Teeth (dental identification)
Superimposition first used in	Ruxton case (1935)
Last bone to fuse	Medial end of clavicle (25 years)
Bertillon system measures	11 body measurements (obsolete)

♥ **Every person is unique — just like every fingerprint, Sammy!**
Forensic science helps speak for those who can no longer speak for themselves.